

DSP Mathematics Reference Notes

Project Vibe-Synth — Internal Technical Document — v1.0 — 2026-04-20

This document collects the core signal-processing mathematics underpinning the algorithms generated by the Vibe-Synth DSP engine. It is intended as a reference for contributors working on the generation layer, BIBO checker, and Faust template library. Equations use discrete-time notation throughout (n denotes the sample index). Continuous-time equivalents are noted where relevant for intuition.

1 Signal Basics and Notation

1.1 Discrete-time signals

A discrete-time signal $x[n]$ is a sequence of real (or complex) numbers indexed by the integer n . In the context of Vibe-Synth, $x[n]$ represents a single audio sample at time $t = n / f_s$, where f_s is the sample rate (default 44 100 Hz).

The unit impulse and unit step:

$$\delta[n] = 1 \text{ if } n = 0, \text{ else } 0$$

$$u[n] = 1 \text{ if } n \geq 0, \text{ else } 0$$

Any FIR filter can be fully characterised by its impulse response $h[n]$, which is the output when $x[n] = \delta[n]$.

1.2 Convolution

Linear time-invariant (LTI) systems produce their output via discrete convolution:

$$y[n] = (x * h)[n] = \sum_k x[k] * h[n - k]$$

For a length- M FIR filter this sum has exactly M non-zero terms.

2 Filter Design

2.1 First-order IIR low-pass filter

The one-pole low-pass filter is the simplest infinite impulse response structure. Its difference equation is:

$$y[n] = (1 - a) * x[n] + a * y[n-1]$$

where the coefficient a determines the cutoff frequency f_c :

$$a = \exp(-2\pi f_c / f_s)$$

This is BIBO stable for $|a| < 1$. In Faust, this is written as:

```
lp1(fc) = _ : *(1 - a) : + ~ *(a) with { a = exp(-2*ma.PI*fc/ma.SR);
};
```

2.2 Resonant low-pass (biquad)

A second-order section (biquad) implements a resonant low-pass filter. Its general difference equation is:

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2]$$

For a resonant low-pass at normalised angular frequency $w_0 = 2\pi f_c / f_s$ and quality factor Q :

Coefficient	Formula
α	$\sin(w_0) / (2Q)$
b_0	$(1 - \cos(w_0)) / 2$
b_1	$1 - \cos(w_0)$
b_2	$(1 - \cos(w_0)) / 2$
a_0	$1 + \alpha$
a_1	$-2 * \cos(w_0)$
a_2	$1 - \alpha$

All coefficients are normalised by a_0 before implementation. Stability requires $|a_2| < 1$ and $|a_1| < 1 + a_2$, guaranteed when $Q > 0$ and $0 < f_c < f_s/2$.

3 Reverberation Algorithms

3.1 Feedback comb filter

A feedback comb filter introduces a delayed copy of the output back into the signal path. Its difference equation is:

$$y[n] = x[n] + g * y[n - M]$$

where M is the delay length in samples and g is the feedback gain. BIBO stability requires $|g| < 1$. The system's frequency response shows peaks (resonances) at multiples of f_s/M .

```
comb(M, g) = _ : (+ ~ (de.delay(8192, M) * g));
```

3.2 Schroeder reverberator

The classic Schroeder reverberator combines parallel feedback comb filters followed by series all-pass filters. The all-pass section has the transfer function:

$$H(z) = (-g + z^M) / (1 - g * z^M)$$

which has unity magnitude response for all frequencies (no colouration) while still introducing time-dispersed echoes. Freeverb, used by Vibe-Synth via `ef.reverb_freeverb`, extends this with tuned comb delays and low-pass damping inside each comb loop to simulate high-frequency air absorption.

4 Waveshaping and Saturation

4.1 Cubic soft clipper

The cubic polynomial waveshaper produces smooth harmonic saturation. For an input x in $[-1, 1]$:

$$y = x - (x^3 / 3)$$

This is the Taylor expansion of $\tanh(x)$ truncated at the third term. The Faust standard library exposes this as `ef.cubicnl(drive, offset)`. The transfer function clips at $|y| = 2/3$ and introduces only odd harmonics (3rd, 5th, ...) for a warm, tube-like character.

4.2 Hard clipping

Hard clipping is the simplest nonlinearity: $y = \text{clip}(x, -A, A)$. It introduces both odd and even harmonics and produces a harsher, transistor-like distortion. In Faust:

```
hardclip(A) = _ : min(A) : max(-A);
```

5 Amplitude and Frequency Modulation

5.1 Tremolo (amplitude modulation)

Tremolo modulates the amplitude of a signal with a low-frequency oscillator (LFO). For a sinusoidal LFO at frequency f_{LFO} and depth d :

$$y[n] = x[n] * (1 - d * (1 - (\sin(2\pi * f_{\text{LFO}} * n / f_s) * 0.5 + 0.5)))$$

When $d = 0$ the signal passes through unchanged. When $d = 1$ the amplitude swings between 0 and 1 (full depth).

5.2 Chorus (pitch modulation via variable delay)

Chorus thickens a signal by adding slightly detuned delayed copies. The delay time is modulated by an LFO:

$$\text{tau}(n) = \text{tau_base} + D * \sin(2\pi * f_{\text{LFO}} * n / f_s)$$

where τ_{base} is the base delay (typically 5-30 ms) and D is the modulation depth in samples. Interpolation between adjacent delay-line taps (linear or Hermite cubic) avoids aliasing artifacts.

6 BIBO Stability Conditions

A system is BIBO (Bounded-Input Bounded-Output) stable if every bounded input produces a bounded output. For linear recursive systems:

FIR filters are unconditionally BIBO stable (finite impulse response implies finite output for any bounded input).

IIR filters are BIBO stable if and only if all poles of their z -domain transfer function $H(z)$ lie strictly inside the unit circle: $|z_k| < 1$ for all poles z_k .

For the feedback comb filter $y[n] = x[n] + g \cdot y[n-M]$, the single pole is at $z = g^{1/M}$. Stability requires $|g| < 1$.

BIBOChecker in `compiler/bibo_checker.py` implements heuristic checks for the most common instability patterns in LLM-generated Faust code: feedback gain ≥ 1.0 , non-productive recursion, and missing damping in reverb feedback paths. These are necessary but not sufficient conditions for BIBO stability.

7 z-Transform Quick Reference

The z-transform converts a discrete-time signal $x[n]$ into a function $X(z)$ of the complex variable z :

$$X(z) = \text{SUM}_{n=-\infty}^{\infty} x[n] * z^{-n}$$

Operation / signal	z-Transform
$x[n - k]$ (delay by k samples)	$z^{-k} * X(z)$
$x[n] + y[n]$	$X(z) + Y(z)$
$a * x[n]$	$a * X(z)$
$\delta[n]$ (unit impulse)	1
$u[n]$ (unit step)	$z / (z - 1)$
$a^n * u[n]$	$z / (z - a), \quad z > a $

7.1 Transfer function and frequency response

For an LTI system described by its difference equation, the z-domain transfer function $H(z) = Y(z)/X(z)$. The frequency response is obtained by evaluating $H(z)$ on the unit circle:

$$H(e^{j*\omega}) = H(z)|_{z = e^{j*\omega}}$$

where $\omega = 2\pi*f/f_s$ is the normalised angular frequency (0 to π for 0 to $f_s/2$). The magnitude response $|H|$ determines the filter's gain at each frequency; the phase response $\arg(H)$ determines group delay.

8 Glossary

BIBO — Bounded-Input Bounded-Output — a stability criterion for LTI systems.

Biquad — A second-order IIR filter section (two poles, two zeros).

Comb filter — A filter whose frequency response resembles the teeth of a comb, produced by adding a delayed copy of the signal.

Difference equation — A recurrence relation expressing the current output sample as a linear combination of past inputs and past outputs.

Faust — A functional programming language for real-time audio signal processing that compiles to C++, WebAssembly, and other targets.

FIR — Finite Impulse Response — a filter whose output depends only on current and past inputs, not past outputs. Always BIBO stable.

IIR — Infinite Impulse Response — a filter with feedback; output depends on past outputs as well as inputs. Stable iff all poles inside the unit circle.

LFO — Low-Frequency Oscillator — a sine or triangle wave below ~20 Hz used for modulation (tremolo, vibrato, chorus).

LTi — Linear Time-Invariant — a system property guaranteeing superposition and shift-invariance; such systems are fully characterised by their impulse response.

Pole — A value of z where the transfer function $H(z)$ goes to infinity. Poles inside the unit circle correspond to decaying exponentials (stable).

Q factor — Quality factor — determines the bandwidth of a resonant filter peak. Higher Q = narrower, sharper peak.

Waveshaper — A memoryless nonlinear function applied to each sample to introduce harmonic distortion.

For questions or corrections contact the generation layer maintainer. This document is auto-generated from the project's internal knowledge base and should be updated whenever a new DSP archetype is added to `dsp_templates.json`.